[1]:
```python
import pandas as pd
import numpy as np
```

[2]:
```python
df = pd.read_csv('/home/subramanyaprasad/Machine Learning/Data Analysis/Notes/03-Pandas/tips.csv')
```

[3]:
```python
df.head()
```

[3]:

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251

Suppose we want to grab the last 4 digits of the CC number:

There's no real built-in function in Pandas which convert integer to String, so use 'apply' to create a custom function.

[12]:
```python
def last_four(num):
    return int(str(num)[-4:])
```

[13]:
```python
df['Last four'] = df['CC Number'].apply(last_four)
```

[14]:
```python
df
```

[14]:

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID	Last four
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959	3410
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608	9230
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458	1322
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260	5994
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251	7221
...	...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657	2842
240	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766	5404
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880	7196
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17	950
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672	8139

244 rows × 12 columns

[15]:
```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   total_bill        244 non-null    float64
 1   tip               244 non-null    float64
 2   sex               244 non-null    str    
 3   smoker            244 non-null    str    
 4   day               244 non-null    str    
 5   time              244 non-null    str    
 6   size              244 non-null    int64  
 7   price_per_person  244 non-null    float64
 8   Payer Name        244 non-null    str    
 9   CC Number         244 non-null    int64  
 10  Payment ID        244 non-null    str    
 11  Last four         244 non-null    int64  
dtypes: float64(3), int64(3), str(6)
memory usage: 23.0 KB
```

[17]:
```python
df['total_bill'].mean()
```

[17]:
```
np.float64(19.78594262295082)
```

[18]:
```python
def yelp(price):
    if price<10:
        return '$'
    elif price >= 10 and price < 30:
    elif price >= 10 and price < 30:
        return '$$'
    else:
        return '$$$'
```

[20]:
```python
df['yelp'] = df['total_bill'].apply(yelp)
df
```

[20]:

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID	Last four	yelp
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959	3410	$$
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608	9230	$$
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458	1322	$$
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260	5994	$$
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251	7221	$$
...	...	...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657	2842	$$
240	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766	5404	$$
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880	7196	$$
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17	950	$$
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672	8139	$$

244 rows × 13 columns

These 'apply' functions must return single value.